

◆ Project Explanation: Facial Emotion Recognition

This project focuses on **facial emotion recognition** using the **icml_face_data dataset**, which contains grayscale facial images labeled with seven emotion classes. The objective was to compare traditional machine learning models with a deep learning approach to identify the most effective method for emotion classification.

Initially, classical machine learning models such as **Decision Tree, Random Forest, and Support Vector Machine (SVM)** were trained using extracted pixel-based features. These models showed strong performance on the training dataset; however, significant overfitting was observed, particularly in Decision Tree and Random Forest models.

To address these limitations, a **deep Convolutional Neural Network (CNN)** was designed to automatically learn spatial and hierarchical features directly from facial images. The CNN architecture consisted of multiple convolutional blocks with batch normalization, max pooling, and dropout layers to improve generalization and reduce overfitting.

The models were evaluated using accuracy on the icml_face_data dataset and validation data to assess both performance and robustness.



One-line version (for viva / interview)

Although traditional models achieved high training accuracy, the CNN demonstrated superior generalization and robustness, making it the most suitable model for facial emotion recognition.